

Film Developer UI MVP (Pico W)

Minimal on-device UI for a film development timer on Raspberry Pi Pico W with:
- 128×64 SH1107 OLED (SPI) - 3 buttons (with internal pull-ups) - DS18B20 temperature probe

Quick Start

```
film-developer/  
  main.py  
  lib/  
    display.py  # 21×8 text grid wrapper  
    sh1107.py   # minimal SH1107 SPI driver  
    input.py    # 3-button debounce/events  
    temp.py     # DS18B20 helper  
    ui.py       # Splash/Main/Menu/Timer stub
```

Run loop entrypoint:

```
python3 film-developer/main.py
```

(On-device you'll deploy these as `main.py` + `lib/` to MicroPython.)

Wiring (Pins)

Follow the full pin map in the ticket reference doc `01-application-architecture-and-system-overview.md`. The essentials are summarized below.

Power (USB)

- Power the Pico W over USB (VBUS → onboard 3V3 regulator). Do not feed 5V into 3V3.
- Power peripherals from 3V3 Out (~300 mA recommended).
- Common GND is required across OLED, DS18B20, and all buttons.

OLED (SH1107, SPI0)

- VCC → 3V3
- GND → GND
- SCK → GP18
- MOSI (DIN) → GP19
- CS → GP17
- DC → GP20
- RST → GP21

Temperature Sensor (DS18B20)

- DATA → GP22 (with 4.7kΩ pull-up to 3V3)

- VCC → 3V3
- GND → GND

Buttons (3×, Internal Pull-Up)

Active-low wiring: each button connects its GPIO to GND when pressed. No external pull-ups needed (enabled in software).

Button	GPIO	Electrical
BTN1	GP14	GP14 → (button) → GND, internal pull-up enabled
BTN2	GP15	GP15 → (button) → GND, internal pull-up enabled
BTN3	GP16	GP16 → (button) → GND, internal pull-up enabled

Notes: - All three buttons share common GND. - Debounce is handled in `lib/input.py` (50 ms default).

Button Roles (MVP)

- BTN1: Start/Select (contextual)
- BTN2: Menu/Navigate (contextual)
- BTN3: Back/Stop (contextual)

Full Wiring Diagram (ASCII)

Orientation: Pico W with micro-USB at the top.

```

USB 5V → VBUS (onboard regulator) → 3V3 Out    > OLED VCC
                                                > DS18B20 VCC
                                                > (optional) other 3V3 peripherals

GND (common)                                > OLED GND
                                                > DS18B20 GND
                                                > BTN1/BTN2/BTN3 to GND when pressed

```

Pico W GPIOs:

```

GP18 (SPI0 SCK)      → OLED SCK
GP19 (SPI0 MOSI)     → OLED DIN
GP17 (SPI0 CSn)      → OLED CS
GP20 (DC)            → OLED DC
GP21 (RST)           → OLED RST

```

```

GP22      → DS18B20 DATA
          [4.7k] → 3V3    (mandatory pull-up)

```

```

GP14 / BTN1 / → GND (internal pull-up enabled in software)

```

GP15 / BTN2 / → GND
 GP16 / BTN3 / → GND

Notes: - Buttons are active-low; internal pull-ups are enabled in `lib/input.py`.
 - DS18B20 requires the 4.7kΩ pull-up from DATA to 3V3.

Pico W Pinout Diagram (ASCII)

USB port at top. Only pins used in this project are labeled with their function.

USB Port

Raspberry Pi Pico W

GP0	1	40	VBUS (5V USB)
GP1	2	39	VSYS
GND	3	38	GND
GP2	4	37	3V3_EN
GP3	5	36	3V3 Out → OLED, DS18B20
GP4	6	35	ADC_VREF
GP5	7	34	GP28
GND	8	33	GND → OLED, DS18B20, BTNs
GP6	9	32	GP27
GP7	10	31	GP26
GP8	11	30	RUN
GP9	12	29	GP22 → DS18B20 DATA (+4.7kΩ)
GND	13	28	GND
GP10/LED1	14	27	GP21 → OLED RST
GP11/LED2	15	26	GP20 → OLED DC
GP12/LED3	16	25	GP19 → OLED DIN (MOSI)
GP13	17	24	GP18 → OLED SCK
GND	18	23	GND
GP14/BTN1	19	22	GP17 → OLED CS
GP15/BTN2	20	21	GP16/BTN3

Legend:

BTN1/2/3: Buttons to GND (internal pull-ups)
 LED1/2/3: LEDs with 220Ω resistors to GND (PWM capable)
 OLED: SH1107 128×64 on SPI0
 DS18B20: Temperature sensor on 1-Wire (GP22 + 4.7kΩ pull-up to 3V3)

Full Pinout (Reference)

- Interactive Pico W pinout: <https://picow.pinout.xyz/>
- Official Pico W A4 pinout PDF: <https://datasheets.raspberrypi.com/picow/PicoW-A4-Pinout.pdf>
 - Use these diagrams to confirm orientation and alternate functions (SPI/I2C/UART/PWM).

References

- Ticket: FILM-DEV-001 (see `reference/01-application-architecture-and-system-overview.md`)
- UI spec: `reference/02-user-interface-specification-and-display-system.md`
- Timer system: `reference/03-timer-system-and-development-process-management.md`